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## Session 3 — Analysis I

### Exercise sheet

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This sheet is about **first-order differential equations**: what they mean, what a solution is, and what the solution looks like. Exercises 1–3 are the toolkit and should take you a few minutes; everything after that is the real subject.

Two rules apply throughout. **Check every solution by substituting it back** into the equation — that is a complete proof that it works, and it costs one line. And whenever you are asked for a solution, **give the interval it is defined on**: a solution is a function together with an interval, not a formula.

#### A. Toolkit (three short exercises)

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**Exercise 1** Differentiate, naming the rule you use.

- (a)  $f(t) = t^2 - 4t + 7$ ;  $g(t) = 5e^{-2t}$ ;  $h(t) = e^{-t^2}$ .
  - (b)  $f(t) = te^{-t}$ . Find where  $f' = 0$ , decide whether it is a maximum or a minimum, and sketch the curve.
  - (c) Someone writes  $\frac{d}{dt}e^{-t/2} = \left(-\frac{t}{2}\right)e^{(-t/2)-1}$ . Which rule have they applied, why does it not apply, and what single question would have prevented the error?
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**Exercise 2** Antiderivatives. Check each answer by differentiating it.

- (a)  $\int_0^2 (3t^2 - 4t + 1) dt$ ;  $\int \frac{2t}{1+t^2} dt$ .
  - (b)  $\int_0^T e^{-t/\tau} dt$ , and its limit as  $T \rightarrow \infty$ . What does the limit say about  $\tau$ ?
  - (c) Why is the condition  $n \neq -1$  needed in  $\int t^n dt = \frac{t^{n+1}}{n+1}$ , and which function fills the gap?
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**Exercise 3** The time constant.

- (a) Write  $e^{-t/5}$  in the form  $e^{at}$  and give  $a$ . For which sign of  $a$  does  $e^{at}$  decay?
- (b) Show that  $R(t) = R_0 e^{-t/\tau}$  satisfies  $R' = -R/\tau$ , and state in words what that equation asserts about forgetting.
- (c) Two learners have  $\tau = 2$  and  $\tau = 8$  days. Without computing anything, say which one forgets faster and why.

#### B. First-order differential equations

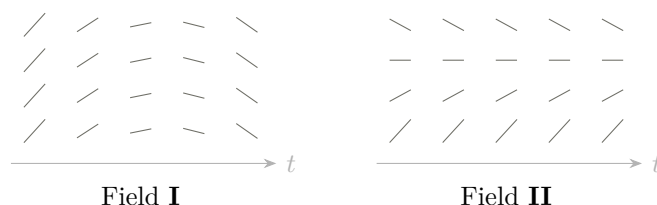
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**Exercise 4** For each equation, say whether it is an *integration problem* ( $\dot{x}$  depends on  $t$  only) or a *differential equation whose right-hand side involves the unknown*, then solve it with  $x(0) = 1$  and give the interval of definition.

- (a)  $\dot{x} = 3t^2$ ; (b)  $\dot{x} = e^{-t}$ ;  
(c)  $\dot{x} = -2x$ ; (d)  $\dot{x} = -2x + 5$ .

**Exercise 5** A slope field *draws, at each point  $(t, x)$ , a short segment of slope  $f(t, x)$* . A *solution* is a curve tangent to the field at every one of its points.

Two fields are drawn below.



- (a) In each field, does the slope depend on  $t$ , on  $x$ , or on both? Justify by what stays constant along which lines.  
(b) One field comes from  $\dot{x} = f(t)$  and the other from  $\dot{x} = f(x)$ . Which is which?  
(c) In each field, sketch the solution through a point near the bottom left and the solution through a point directly above it. How are the two curves related in field I? In field II?

**Exercise 6** Someone proposes  $\varphi(x) = -x^2/2 + 3x$  as the solution of  $\dot{x} = -(x - 3)$ .

- (a) Give the one-line *type* argument showing it cannot be a solution.  
(b) Differentiate  $t \mapsto \varphi(x(t))$  with the chain rule and show the equation is not satisfied.  
(c) Give the correct solution with  $x(0) = x_0$ , and verify it by substitution.

**Exercise 7** The linear equation  $\dot{x} = ax + b$ , with  $a \neq 0$  and  $x^* = -b/a$ .

- (a) Verify by substitution that  $x(t) = x^* + (x_0 - x^*)e^{at}$  is a solution with  $x(0) = x_0$ , and say on which interval.  
(b)  $\dot{x} = -x + 4$ : give  $x^*$  and  $\tau$ , decide stability, and describe the trajectory from  $x_0 = 10$  and from  $x_0 = 0$ .  
(c)  $\dot{x} = 3x - 6$ : give  $x^*$ , decide stability, and describe what happens from  $x_0 = 1.9$  and from  $x_0 = 2.1$ .  
(d) What goes wrong if  $a = 0$ ? Solve  $\dot{x} = b$  directly and say why there is no equilibrium.

**Exercise 8** *Cauchy–Lipschitz: if  $f$  is continuous and locally Lipschitz in  $x$  — in particular if  $f$  is  $C^1$  — then through each point there passes exactly one maximal solution.*

Consider  $\dot{x} = 2\sqrt{|x|}$ .

- (a) Does the Cauchy problem with  $x(0) = 1$  have a unique solution near  $t = 0$ ? Why?
  - (b) Check that  $x \equiv 0$  and  $x(t) = t^2$  (for  $t \geq 0$ ) both solve the problem with  $x(0) = 0$ . Which hypothesis fails, and at which point?
  - (c) Uniqueness implies that two distinct integral curves never meet. Use this — without solving anything — to explain why the solution of  $\dot{x} = -x + 4$  starting at  $x_0 = 10$  never reaches the value 4.
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**Exercise 9** Maximal solutions of  $\dot{x} = x^2$ .

- (a) Verify that  $x(t) = \frac{1}{1-t}$  solves the problem with  $x(0) = 1$ , and give the largest interval containing 0 on which it is a solution. Why is the answer not  $\mathbb{R} \setminus \{1\}$ ?
  - (b) Verify that  $x(t) = \frac{-1}{1+t}$  solves the problem with  $x(0) = -1$ , and give its maximal interval.
  - (c) The function  $f(x) = x^2$  is  $C^\infty$  on all of  $\mathbb{R}$ . Explain, in one sentence, why that does not prevent (a) from ceasing to exist at  $t = 1$ .
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**Exercise 10** Phase lines. For an autonomous equation  $\dot{x} = f(x)$ : plot  $f$ , mark its zeros, and put an arrow pointing right where  $f > 0$  and left where  $f < 0$ .

- (a) Draw the phase lines of  $\dot{x} = -x + 3$  and  $\dot{x} = x - 3$ . Which equilibrium is stable, and why, *without* solving either equation?
  - (b)  $\dot{x} = x(1-x)$ : find the equilibria, classify them using the sign of  $f'$  at each, and draw the phase line. Where does a solution starting at  $x_0 = 0.1$  go? At  $x_0 = -0.1$ ?
  - (c)  $\dot{x} = x^2$ : show that the criterion on  $f'$  says nothing at the equilibrium, and settle the question with the phase line instead.
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**Exercise 11** Euler's method:  $x_{n+1} = x_n + \Delta t f(t_n, x_n)$ .

- (a) Take two steps of  $\dot{x} = 0.1(100-x)$  from  $x_0 = 20$  with  $\Delta t = 1$ , and compare with the exact value  $x(2) = 100 - 80e^{-0.2}$ .
- (b) Is the numerical value above or below the exact one? Explain using the slope field.
- (c) For  $\dot{x} = ax$ , show that  $x_n = (1 + a\Delta t)^n x_0$ , and deduce the largest step that does not blow up when  $a = -1/5$ .
- (d) In one sentence: why is “the simulation did not explode” not the same as “the simulation is accurate”?